



AuthiChain Production Deployment Guide

Complete Guide to Deploying AuthiChain NFT Marketplace to Production

Version: 1.0.0

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Status: Ready for Production Deployment



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Overview

This documentation package provides everything you need to deploy AuthiChain to production. The platform includes:

Core Features

-  **5-Tier Subscription System** (Starter, Professional, Business, Enterprise, White Label)
-  **3 Billing Cycles** per tier (Monthly, Annual, Lifetime)
-  **NFT Minting & Management** with IPFS storage
-  **Stripe Payment Integration** for subscriptions
-  **PostgreSQL Database** with Prisma ORM
-  **NextAuth Authentication** with session management
-  **Responsive Next.js Frontend** (React 18)
-  **RESTful API** for integrations
-  **MetaMask Wallet Integration** (optional)

Production-Ready Features

-  **SSL/HTTPS support**
-  **Database migrations**
-  **Error tracking and monitoring**
-  **Automated backups**
-  **Rate limiting**
-  **Security headers**

-  CORS configuration
 -  Email notifications
 -  Analytics integration
-

Documentation Index

1. Setup & Configuration

PRODUCTION_SETUP.md (./PRODUCTION_SETUP.md)

The Master Setup Guide

- NFT.Storage configuration (IPFS storage)
- Stripe Live Keys setup
- Database setup (Railway, Supabase, AWS RDS)
- Environment variable configuration
- Security best practices
- Testing procedures

Start here first!

API_KEYS_SETUP.md (./API_KEYS_SETUP.md)

How to Obtain All API Keys

- NFT.Storage API key (FREE)
- Stripe Live API keys
- Stripe Webhook secret
- NextAuth secret generation
- Database connection URLs

Estimated time: 30 minutes

STRIPE_PRICE_SETUP.md (./STRIPE_PRICE_SETUP.md)

Creating 15 Stripe Price IDs

- Step-by-step Stripe product creation
- All 15 price configurations
- Price ID tracking spreadsheet
- Testing checkout flow

Critical for payment processing!

2. Deployment Guides

Choose your deployment platform:

DEPLOYMENT_VERCEL.md (./DEPLOYMENT_VERCEL.md)

Deploy to Vercel (Recommended for Beginners)

-  Easiest deployment

-  Optimized for Next.js
-  Auto-scaling
-  Free SSL
-  Global CDN

Best for: Serverless deployment, global reach, minimal ops

DEPLOYMENT_RAILWAY.md (./DEPLOYMENT_RAILWAY.md)

Deploy to Railway (Recommended for Simplicity)

-  One-click PostgreSQL
-  No cold starts
-  Simple pricing
-  Great DX

Best for: Fast deployment, integrated database, predictable costs

DEPLOYMENT_AWS.md (./DEPLOYMENT_AWS.md)

Deploy to AWS (Recommended for Enterprise)

-  Full control
-  Enterprise-grade
-  Compliance-ready
-  Scalable to millions

Best for: Large scale, compliance requirements, AWS ecosystem

3. Environment Configuration

.env.example (./app/.env.example)

Development Environment Template

- Comprehensive variable documentation
- Example values for all services
- Quick start guide included
- Security warnings

Use for local development

.env.production.example (./app/.env.production.example)

Production Environment Template

- Production-specific variables
- Live Stripe keys template
- Security checklist
- Feature flags

Use for production deployment

4. Pre-Launch

PRODUCTION_CHECKLIST.md (./PRODUCTION_CHECKLIST.md)

Complete Pre-Launch Checklist

- 23 sections covering everything
- Security testing
- Performance testing
- Legal compliance
- Post-launch monitoring

Complete before going live!



Quick Start Path

Follow this path for fastest deployment:

Path A: Beginners (Vercel + Railway)

Estimated Time: 2-3 hours

1. ☒ **Read [PRODUCTION_SETUP.md](#)** (./PRODUCTION_SETUP.md) (30 min)
2. ☒ **Get API Keys** using [API_KEYS_SETUP.md](#) (./API_KEYS_SETUP.md) (30 min)
3. ☒ **Create Stripe Prices** using [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md) (45 min)
4. ☒ **Deploy to Vercel** using [DEPLOYMENT_VERCEL.md](#) (./DEPLOYMENT_VERCEL.md) (45 min)
5. ☒ **Complete Checklist** using [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md) (2-3 hours)

Total: ~5-6 hours for first deployment

Path B: Intermediate (Railway)

Estimated Time: 2-4 hours

1. ☒ **Read [PRODUCTION_SETUP.md](#)** (./PRODUCTION_SETUP.md) (30 min)
2. ☒ **Get API Keys** using [API_KEYS_SETUP.md](#) (./API_KEYS_SETUP.md) (30 min)
3. ☒ **Create Stripe Prices** using [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md) (45 min)
4. ☒ **Deploy to Railway** using [DEPLOYMENT_RAILWAY.md](#) (./DEPLOYMENT_RAILWAY.md) (1 hour)
5. ☒ **Complete Checklist** using [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md) (2-3 hours)

Total: ~5-7 hours for first deployment

Path C: Advanced (AWS)

Estimated Time: 4-6 hours

1. ☒ **Read [PRODUCTION_SETUP.md](#)** (./PRODUCTION_SETUP.md) (30 min)
2. ☒ **Get API Keys** using [API_KEYS_SETUP.md](#) (./API_KEYS_SETUP.md) (30 min)
3. ☒ **Create Stripe Prices** using [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md) (45 min)

4.  **Deploy to AWS** using [DEPLOYMENT_AWS.md](#) (./DEPLOYMENT_AWS.md) (2-3 hours)
5.  **Complete Checklist** using [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md) (2-3 hours)

Total: ~6-8 hours for first deployment

Required Services

Free Services ✨

1. **NFT.Storage** (IPFS Storage)
 - Cost: FREE forever
 - Storage: Unlimited
 - Bandwidth: Unlimited
 - Sign up: <https://nft.storage>
2. **GitHub** (Code Repository)
 - Cost: FREE
 - Private repos included
 - Sign up: <https://github.com>

Paid Services 💳

Essential Services

1. **Stripe** (Payment Processing)
 - Cost: FREE (2.9% + \$0.30 per transaction)
 - No monthly fees
 - Live keys require activated account
 - Sign up: <https://stripe.com>

2. PostgreSQL Database

Railway:

- Cost: \$5/month (after free trial)
- 500 execution hours
- Easy setup
- Sign up: <https://railway.app>

Supabase:

- Cost: FREE tier available
- Pro: \$25/month
- Advanced features
- Sign up: <https://supabase.com>

AWS RDS:

- Cost: ~\$15/month (db.t3.micro)
- Enterprise-grade

- Full control
- Sign up: <https://aws.amazon.com>

1. Hosting Platform

Vercel:

- Cost: FREE hobby tier
- Pro: \$20/month (recommended for production)
- Sign up: <https://vercel.com>

Railway:

- Cost: \$5-20/month
- Includes database
- Sign up: <https://railway.app>

AWS:

- Cost: ~\$50-200/month (varies)
 - Full infrastructure
 - Sign up: <https://aws.amazon.com>
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Optional Services

1. Domain Name

- Cost: \$10-15/year
- Providers: Namecheap, GoDaddy, Google Domains

2. Email Service (for notifications)

- SendGrid: FREE (100 emails/day)
- Postmark: \$10/month (10,000 emails)
- AWS SES: \$0.10 per 1,000 emails

3. Error Tracking

- Sentry: FREE tier available
- Pro: \$26/month
- Sign up: <https://sentry.io>

4. Analytics

- Google Analytics: FREE
 - PostHog: FREE tier available
 - Sign up: <https://analytics.google.com>
-

Estimated Costs

Minimal Production Setup

Best for: MVP, testing, small user base

Service	Provider	Cost
Hosting	Vercel Hobby	FREE
Database	Railway	\$5/mo
Payment	Stripe	2.9% + \$0.30
IPFS Storage	NFT.Storage	FREE
Domain	Namecheap	\$12/yr
Total		~\$6/month

Standard Production Setup

Best for: Growing startups, production apps

Service	Provider	Cost
Hosting	Vercel Pro	\$20/mo
Database	Railway	\$20/mo
Payment	Stripe	2.9% + \$0.30
IPFS Storage	NFT.Storage	FREE
Domain	Namecheap	\$12/yr
Error Tracking	Sentry	\$26/mo
Email	SendGrid	\$15/mo
Total		~\$82/month

Enterprise Production Setup

Best for: Large scale, compliance, high traffic

Service	Provider	Cost
Hosting	AWS (EC2, ALB, CloudFront)	\$150/mo
Database	AWS RDS (Multi-AZ)	\$100/mo
Payment	Stripe	2.9% + \$0.30
IPFS Storage	NFT.Storage	FREE
Domain	Premium .com	\$12/yr
Error Tracking	Sentry Business	\$80/mo
Email	AWS SES	\$10/mo
Monitoring	CloudWatch	\$30/mo
Backups	AWS S3	\$20/mo
Total		~\$390/month

Note: Payment processing costs (2.9% + \$0.30) are per transaction. For \$10,000/mo in revenue, that's ~\$290 in Stripe fees.



Deployment Timeline

First-Time Deployment

Estimated Total: 1-2 days

Day 1: Setup & Configuration (4-6 hours)

- ☐ Read documentation (1 hour)
- ☐ Obtain API keys (1 hour)
- ☐ Create Stripe products and prices (1 hour)
- ☐ Set up database (30 min)
- ☐ Configure environment variables (30 min)
- ☐ Deploy application (1 hour)
- ☐ Configure domain and SSL (30 min)

Day 2: Testing & Launch (4-6 hours)

- ☐ Run functional tests (2 hours)
- ☐ Performance testing (1 hour)
- ☐ Security audit (1 hour)
- ☐ Complete pre-launch checklist (2 hours)
- ☐ Go live! (30 min)
- ☐ Monitor first few hours (ongoing)

Subsequent Deployments

Estimated: 30-60 minutes

With all services already configured:

- [] Update code (10 min)
 - [] Run tests (10 min)
 - [] Deploy (5 min)
 - [] Verify deployment (15 min)
-

Learning Path

Prerequisites Knowledge

Required:

- Basic command line usage
- Git basics (clone, commit, push)
- Environment variables concept

Helpful but not required:

- Next.js/React knowledge
 - PostgreSQL basics
 - Stripe API familiarity
-

Documentation Reading Order

1. **Start Here:**

- [PRODUCTION_SETUP.md](#) (./PRODUCTION_SETUP.md) - Read completely

2. **Get API Keys:**

- [API_KEYS_SETUP.md](#) (./API_KEYS_SETUP.md) - Follow step-by-step

3. **Configure Payments:**

- [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md) - Create all 15 prices

4. **Choose Deployment:**

- [DEPLOYMENT_VERCEL.md](#) (./DEPLOYMENT_VERCEL.md) (easiest)
- OR [DEPLOYMENT_RAILWAY.md](#) (./DEPLOYMENT_RAILWAY.md) (simple)
- OR [DEPLOYMENT_AWS.md](#) (./DEPLOYMENT_AWS.md) (enterprise)

5. **Final Checks:**

- [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md) - Complete before launch
-



Troubleshooting

Common Issues

“Build Failed”

Solution: Check [DEPLOYMENT_VERCEL.md](#) (./DEPLOYMENT_VERCEL.md#troubleshooting) section

“Database Connection Timeout”

Solution: Check [PRODUCTION_SETUP.md](#) (./PRODUCTION_SETUP.md#troubleshooting) section

“Stripe Webhook Not Working”

Solution: Check [STRIPE_PRICE_SETUP.md](#) (./STRIPE_PRICE_SETUP.md#common-issues) section

“Environment Variable Undefined”

Solution: Verify all required variables in [.env.production.example](#) (./app/.env.production.example)

Support Resources

Documentation

- All guides include troubleshooting sections
- Each guide has specific examples
- Screenshots included where helpful

External Resources

- [Next.js Documentation](https://nextjs.org/docs) (https://nextjs.org/docs)
- [Prisma Documentation](https://www.prisma.io/docs) (https://www.prisma.io/docs)
- [Stripe Documentation](https://stripe.com/docs) (https://stripe.com/docs)
- [NFT.Storage Documentation](https://nft.storage/docs) (https://nft.storage/docs)

Community

- Stack Overflow (tag: `nextjs` , `prisma` , `stripe`)
- Vercel Discord (for Vercel deployments)
- Railway Discord (for Railway deployments)

Success Criteria

Your deployment is successful when:

- [x] Application accessible via HTTPS
 - [x] Users can register and log in
 - [x] Users can subscribe to a tier
 - [x] Payment processing works
 - [x] NFTs can be minted
 - [x] Images uploaded to IPFS successfully
 - [x] Database persists data
 - [x] Webhooks process correctly
 - [x] No critical errors in logs
 - [x] All security checks pass
-

Launch Preparation

Before announcing your launch:

1. ☒ **Complete `PRODUCTION_CHECKLIST.md`** (`./PRODUCTION_CHECKLIST.md`)
 2. ☒ **Test with real payment** (will charge, cancel after)
 3. ☒ **Monitor for 24 hours** before public announcement
 4. ☒ **Set up error tracking** and alerts
 5. ☒ **Prepare support channels** (email, chat, etc.)
 6. ☒ **Document known issues** (if any)
 7. ☒ **Have rollback plan** ready
-

Post-Launch

First 24 Hours

- Monitor error rates
- Watch server metrics
- Respond to user issues
- Track conversion rates

First Week

- Analyze user behavior
- Optimize bottlenecks
- Address feedback
- Plan improvements

Ongoing

- Weekly error log review
 - Monthly security updates
 - Quarterly feature releases
 - Annual security audit
-

Update Process

To update AuthiChain after initial deployment:

1. Local Testing:

```
bash
git pull origin main
npm install
npm run build
npm test
```

2. Deploy Update:

- Vercel: `git push` (auto-deploys)

- Railway: `git push` (auto-deploys)
- AWS: Follow [DEPLOYMENT_AWS.md](#) (`./DEPLOYMENT_AWS.md#updates`)

3. **Verify Deployment:**

- Test critical paths
- Check error rates
- Monitor for 1 hour



Configuration Checklist

Quick reference for required configuration:

API Keys (5 required)

- ☐ NFT.Storage API Key
- ☐ Stripe Secret Key
- ☐ Stripe Publishable Key
- ☐ Stripe Webhook Secret
- ☐ NextAuth Secret

Price IDs (15 required)

- ☐ Starter: Monthly, Annual, Lifetime
- ☐ Professional: Monthly, Annual, Lifetime
- ☐ Business: Monthly, Annual, Lifetime
- ☐ Enterprise: Monthly, Annual, Lifetime
- ☐ White Label: Monthly, Annual, Lifetime

Infrastructure

- ☐ PostgreSQL Database
- ☐ Hosting Platform
- ☐ Domain Name
- ☐ SSL Certificate

Optional but Recommended

- ☐ Error Tracking (Sentry)
- ☐ Analytics (Google Analytics)
- ☐ Email Service (SendGrid)
- ☐ Uptime Monitoring



Best Practices

Security

1. Rotate secrets every 90 days
2. Use different secrets for dev/staging/production
3. Enable 2FA on all service accounts
4. Review access logs monthly

5. Keep dependencies updated

Performance

1. Enable caching where possible
2. Optimize database queries
3. Use CDN for static assets
4. Monitor response times
5. Scale proactively

Reliability

1. Enable database backups
2. Set up uptime monitoring
3. Configure alerting
4. Document runbooks
5. Test disaster recovery



Maintenance Schedule

Daily

- Monitor error rates
- Check payment processing
- Review user feedback

Weekly

- Review error logs
- Check database performance
- Analyze user metrics

Monthly

- Update dependencies
- Review security patches
- Optimize performance
- Database maintenance

Quarterly

- Rotate API keys
 - Security audit
 - Disaster recovery test
 - Team access review
-

Next Steps After Deployment

Once deployed, consider:

1. Marketing:

- SEO optimization
- Social media presence
- Content marketing
- Paid advertising

2. Features:

- User-requested features
- A/B testing
- Analytics insights
- Performance improvements

3. Scaling:

- Auto-scaling configuration
- Database optimization
- Caching strategies
- Load balancing

4. Business:

- Customer support system
- Billing optimization
- Refund policy
- Churn analysis

Document Versions

Document	Version	Last Updated
PRODUCTION_SETUP.md	1.0.0	Oct 19, 2025
API_KEYS_SETUP.md	1.0.0	Oct 19, 2025
STRIPE_PRICE_SETUP.md	1.0.0	Oct 19, 2025
DEPLOYMENT_VERCEL.md	1.0.0	Oct 19, 2025
DEPLOYMENT_RAILWAY.md	1.0.0	Oct 19, 2025
DEPLOYMENT_AWS.md	1.0.0	Oct 19, 2025
PRODUCTION_CHECKLIST.md	1.0.0	Oct 19, 2025
.env.example	1.0.0	Oct 19, 2025
.env.production.example	1.0.0	Oct 19, 2025

Final Notes

This documentation package represents a complete production deployment guide for AuthiChain. Every document has been carefully crafted with:

-  Step-by-step instructions
-  Real-world examples
-  Troubleshooting sections
-  Security best practices
-  Time estimates
-  Cost breakdowns

You have everything you need to launch AuthiChain successfully!

Questions?

If you encounter any issues:

1. Check the specific guide's troubleshooting section
 2. Review [PRODUCTION_CHECKLIST.md](#) (./PRODUCTION_CHECKLIST.md)
 3. Consult external documentation (Stripe, Vercel, etc.)
 4. Search Stack Overflow for similar issues
-

Ready to deploy? Start with [PRODUCTION_SETUP.md](#) (./PRODUCTION_SETUP.md)!

Good luck with your AuthiChain launch! 